

Phase 2: Accident Cause Code EDA

This notebook continues the analysis started in `phase-1-proposal.ipynb`. Phase 1 established that metro and rural counties have fundamentally different accident patterns - both in frequency and cause distribution.

The goal here is to identify the **top 5 features** that influence accident cause codes (H, M, T, E, S) through exploratory data analysis. This work supports the classification track of our Phase 2 parallel analysis, alongside Tim and Matt's damage cost investigation.

Setup & Data Loading

Loading the processed merged dataframe from Phase 1. This is the cleaned dataset saved from V2 after the FRA accident data was merged with USDA county-level population and rural-urban classifications.

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.style as style
from sklearn.preprocessing import LabelEncoder
from scipy.stats import pearsonr, chi2_contingency

RANDOM_SEED = 521
np.random.seed(RANDOM_SEED)

sns.set_theme(style="whitegrid")
style.use('tableau-colorblind10')

# Load processed merged dataframe
df_merged = pd.read_csv("../data/processed/df_merged.csv.zip")

# Quick check
print(f"Shape: {df_merged.shape}")
df_merged.head()
```

Shape: (34562, 173)

```
/var/folders/h7/j749vy8n45n8p147xy2ll3vw0000gn/T/ipykernel_34241/4133010811.py:15: DtypeWarning: Columns (0: Persons Evacuated, 1: Division Code, 2: Tra
in Direction Code, 3: Remote Control Locomotive Code, 4: First Car Number) h
ave mixed types. Specify dtype option on import or set low_memory=False.
df_merged = pd.read_csv("../data/processed/df_merged.csv.zip")
```

Out[]:

	Reporting Railroad Code	Reporting Railroad Name	Year	Accident Number	PDF L
0	BNSF	BNSF Railway Company	2013	NW0813104	https://safetydata.fra.dot.gov/Officeofsafet
1	CSX	CSX Transportation	2014	000131012	https://safetydata.fra.dot.gov/Officeofsafet
2	CSX	CSX Transportation	2020	000191201	https://safetydata.fra.dot.gov/Officeofsafet
3	CSX	CSX Transportation	2013	000121335	https://safetydata.fra.dot.gov/Officeofsafet
4	SFRV	South Florida Regional Transportation Authority	2024	03172024	https://safetydata.fra.dot.gov/Officeofsafet

Recreating Cause Category

The `Cause_Category` column wasn't saved in the processed CSV - it was created in V2 by grabbing the first letter of `Primary Accident Cause Code`. Recreating it here so we can work with it.

```
In [132... # Recreate Cause_Category from Primary Accident Cause Code
df_merged['Cause_Category'] = df_merged['Primary Accident Cause Code'].str[0]

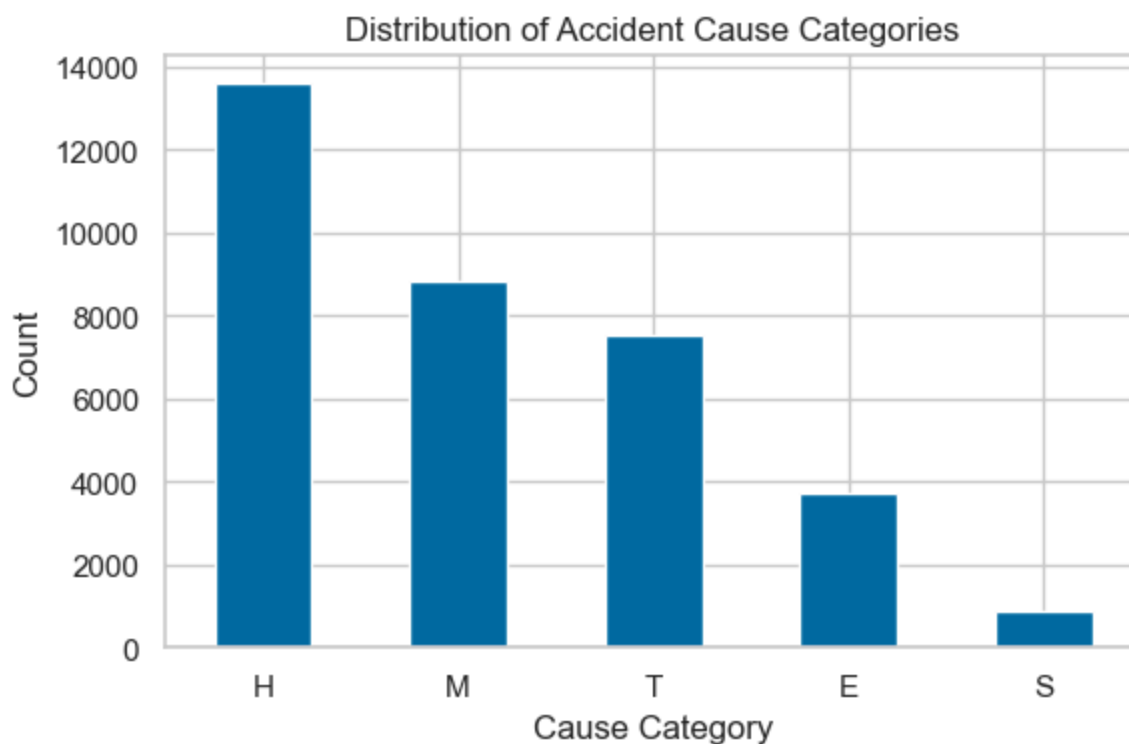
# Now check the distribution
print(df_merged['Cause_Category'].value_counts())
```

```
Cause_Category
H      13582
M      8856
T      7534
E      3721
S       867
Name: count, dtype: int64
```

Cause Category Distribution

Visualizing our target variable. Important to see the balance across categories before moving into feature exploration - imbalanced classes can affect classification later.

```
In [133... # Visualize cause category distribution
plt.figure(figsize=(6, 4))
df_merged['Cause_Category'].value_counts().plot(kind='bar')
plt.title('Distribution of Accident Cause Categories')
plt.xlabel('Cause Category')
plt.ylabel('Count')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



Exploring Categorical Features

First, let's see what string columns we're working with before deciding which ones are worth converting to numerical values.

```
In [134... pd.set_option('display.max_columns', None)
df_merged.columns.tolist()
```

```
Out[134... ['Reporting Railroad Code',
            'Reporting Railroad Name',
            'Year',
            'Accident Number',
            'PDF Link',
            'Accident Year',
            'Accident Month',
            'Other Railroad Code',
            'Other Railroad Name',
            'Other Accident Number',
            'Other Accident Year',
            'Other Accident Month',
            'Maintenance Railroad Code',
            'Maintenance Railroad Name',
            'Maintenance Accident Number',
            'Maintenance Accident Year',
            'Maintenance Accident Month',
            'Grade Crossing ID',
            'Day',
            'Date',
            'Time',
            'Accident Type Code',
            'Accident Type',
            'Hazmat Cars',
            'Hazmat Cars Damaged',
            'Hazmat Released Cars',
            'Persons Evacuated',
            'Subdivision',
            'Division Code',
            'Division',
            'Station',
            'Milepost',
            'State Code',
            'State Abbreviation',
            'State Name',
            'County Code',
            'County Name',
            'District',
            'Temperature',
            'Visibility Code',
            'Visibility',
            'Weather Condition Code',
            'Weather Condition',
            'Track Type Code',
            'Track Type',
            'Track Name',
            'Track Class',
            'Track Density',
            'Train Direction Code',
            'Train Direction',
            'Equipment Type Code',
            'Equipment Type',
            'Equipment Attended',
            'Train Number',
            'Train Speed',
            'Recorded Estimated Speed',
```

'Maximum Speed',
'Gross Tonnage',
'Signalization Code',
'Signalization',
'Method of Operation Code',
'Method of Operation',
'Adjunct Code 1',
'Adjunct Name 1',
'Adjunct Code 2',
'Adjunct Name 2',
'Adjunct Code 3',
'Adjunct Name 3',
'Remote Control Locomotive Code',
'Remote Control Locomotive',
'First Car Initials',
'First Car Number',
'First Car Position',
'First Car Loaded',
'Causing Car Initials',
'Causing Car Number',
'Causing Car Position',
'Causing Car Loaded',
'Positive Alcohol Tests',
'Positive Drug Tests',
'Passengers Transported',
'Head End Locomotives',
'Mid Train Manual Locomotives',
'Mid Train Remote Locomotives',
'Rear End Manual Locomotives',
'Rear End Remote Locomotives',
'Derailed Head End Locomotives',
'Derailed Mid Train Manual Locomotives',
'Derailed Mid Train Remote Locomotives',
'Derailed Rear End Manual Locomotives',
'Derailed Rear End Remote Locomotives',
'Loaded Freight Cars',
'Loaded Passenger Cars',
'Empty Freight Cars',
'Empty Passenger Cars',
'Cabooses',
'Derailed Loaded Freight Cars',
'Derailed Loaded Passenger Cars',
'Derailed Empty Freight Cars',
'Derailed Empty Passenger Cars',
'Derailed Cabooses',
'Equipment Damage Cost',
'Track Damage Cost',
'Total Damage Cost',
'Primary Accident Cause Code',
'Primary Accident Cause',
'Contributing Accident Cause Code',
'Contributing Accident Cause',
'Accident Cause Code',
'Accident Cause',
'Engineers On Duty',
'Firemen On Duty',

'Conductors On Duty',
'Brakemen On Duty',
'Hours Engineers On Duty',
'Minutes Engineers On Duty',
'Hours Conductors On Duty',
'Minutes Conductors On Duty',
'Railroad Employees Killed',
'Railroad Employees Injured',
'Passengers Killed',
'Passengers Injured',
'Others Killed',
'Others Injured',
'Reporting Railroad Fatalities for 55a',
'Reporting Railroad Injuries for 55a',
'Total Persons Killed',
'Total Persons Injured',
'Total Killed Form 54',
'Total Injured Form 54',
'Special Study 1',
'Special Study 2',
'Latitude',
'Longitude',
'Narrative',
'Joint Track Type',
'Joint Track Class',
'Class Code',
'Class',
'Joint CD',
'Incident Key',
'Report Key',
'Reporting Railroad Class',
'Reporting Railroad SMT Grouping',
'Reporting Parent Railroad Code',
'Reporting Parent Railroad Name',
'Reporting Railroad Holding Company',
'Reporting Railroad Individual Class',
'Reporting Railroad Passenger',
'Reporting Railroad Commuter',
'Reporting Railroad Switching Terminal',
'Reporting Railroad Tourist',
'Reporting Railroad Freight',
'Reporting Railroad Short Line',
'Location',
'State Code FIPS',
'County Code FIPS',
'FIPS',
'Description',
'Population_2020',
'RUCC_2023',
'Cause_Category',
'total_cars',
'empty_loaded_ratio',
'pct_empty',
'tonnage_per_car',
'Equipment_Type_Mapped',
'Day_of_Week',

```
'Hour',  
'Time_of_Day',  
'Time_parsed',  
'Region',  
'Cause_Name']
```

Identifying Candidate Features for Encoding

Scanning categorical columns to see which ones have a manageable number of unique values - good candidates for label encoding.

```
In [135... # View a few candidates up close  
candidates = ['Accident Type', 'Weather Condition', 'Track Type', 'Visibilit  
  
for col in candidates:  
    print(f"\n{col} - unique values: {df_merged[col].nunique()}")  
    print(df_merged[col].value_counts().head())
```



```
Accident Type – unique values: 13
Accident Type
Derailment          19113
Other impacts       5114
Hwy–rail crossing   3294
Other (describe in narrative) 2155
Side collision      1689
Name: count, dtype: int64
```

```
Weather Condition – unique values: 6
Weather Condition
Clear      23402
Cloudy     7692
Rain       2422
Snow       643
Fog        324
Name: count, dtype: int64
```

```
Track Type – unique values: 4
Track Type
Yard      19236
Main      11644
Industry   2893
Siding     787
Name: count, dtype: int64
```

```
Visibility – unique values: 4
Visibility
Day      15515
Dark     12227
Dawn      3512
Dusk      3300
Name: count, dtype: int64
```

```
State Abbreviation – unique values: 49
State Abbreviation
TX      4220
IL      2816
CA      2130
OH      1334
IN      1308
Name: count, dtype: int64
```

Checking Visibility

Quick look at Visibility - unique values and distribution before encoding.

```
In [136... print(f"Visibility – unique values: {df_merged['Visibility'].nunique()}")
print(df_merged['Visibility'].value_counts())
```

Visibility – unique values: 4
Visibility
Day 15515
Dark 12227
Dawn 3512
Dusk 3300
Name: count, dtype: int64

Feature Engineering: US Census Regions

```
In [137... # Create a dictionary mapping State Abbreviations to US Census Regions
state_to_region = {
    # Northeast
    'CT': 'Northeast', 'ME': 'Northeast', 'MA': 'Northeast', 'NH': 'Northeast',
    'RI': 'Northeast', 'VT': 'Northeast', 'NJ': 'Northeast', 'NY': 'Northeast',
    'PA': 'Northeast',

    # Midwest
    'IL': 'Midwest', 'IN': 'Midwest', 'MI': 'Midwest', 'OH': 'Midwest',
    'WI': 'Midwest', 'IA': 'Midwest', 'KS': 'Midwest', 'MN': 'Midwest',
    'MO': 'Midwest', 'NE': 'Midwest', 'ND': 'Midwest', 'SD': 'Midwest',

    # South
    'DE': 'South', 'DC': 'South', 'FL': 'South', 'GA': 'South', 'MD': 'South',
    'NC': 'South', 'SC': 'South', 'VA': 'South', 'WV': 'South', 'AL': 'South',
    'KY': 'South', 'MS': 'South', 'TN': 'South', 'AR': 'South', 'LA': 'South',
    'OK': 'South', 'TX': 'South',

    # West
    'AZ': 'West', 'CO': 'West', 'ID': 'West', 'MT': 'West', 'NV': 'West',
    'NM': 'West', 'UT': 'West', 'WY': 'West', 'AK': 'West', 'CA': 'West',
    'HI': 'West', 'OR': 'West', 'WA': 'West'
}

# Map the regions to a new column
df_merged['Region'] = df_merged['State Abbreviation'].map(state_to_region)

# Verify the new column
print(df_merged[['State Abbreviation', 'Region']].head())
print("\nRegion Counts:")
print(df_merged['Region'].value_counts(dropna=False))
```

	State	Abbreviation	Region
0	WA	West	
1	KY	South	
2	TN	South	
3	SC	South	
4	FL	South	

Region Counts:

Region	count
South	13818
Midwest	11218
West	5963
Northeast	3563

Name: count, dtype: int64

```
In [138... # Ensure Cause_Category exists by extracting the first letter of the Primary
# (H=Human Factor, M=Miscellaneous, T=Track, E=Equipment, S=Signal)
if 'Cause_Category' not in df_merged.columns:
    df_merged['Cause_Category'] = df_merged['Primary Accident Cause Code'].s

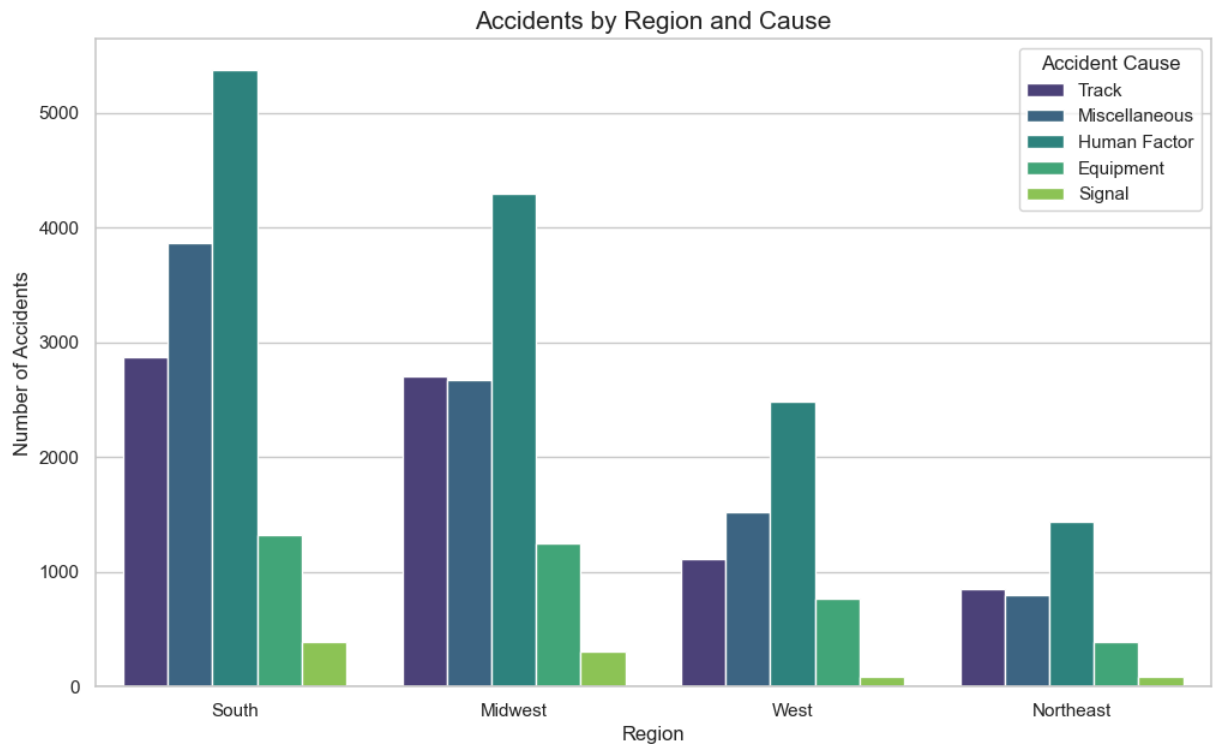
# Map the single-letter codes to full names for the legend
cause_map = {
    'T': 'Track',
    'E': 'Equipment',
    'H': 'Human Factor',
    'S': 'Signal',
    'M': 'Miscellaneous'
}
df_merged['Cause_Name'] = df_merged['Cause_Category'].map(cause_map)

# Set the figure size
plt.figure(figsize=(12, 7))

# Create a count plot with Region on x-axis and hue (grouping) by Cause
sns.countplot(
    data=df_merged,
    x='Region',
    hue='Cause_Name',
    order=df_merged['Region'].value_counts().index,
    palette='viridis'
)

# Add labels and title
plt.title('Accidents by Region and Cause', fontsize=15)
plt.xlabel('Region', fontsize=12)
plt.ylabel('Number of Accidents', fontsize=12)
plt.legend(title='Accident Cause')

# Show the plot
plt.show()
```

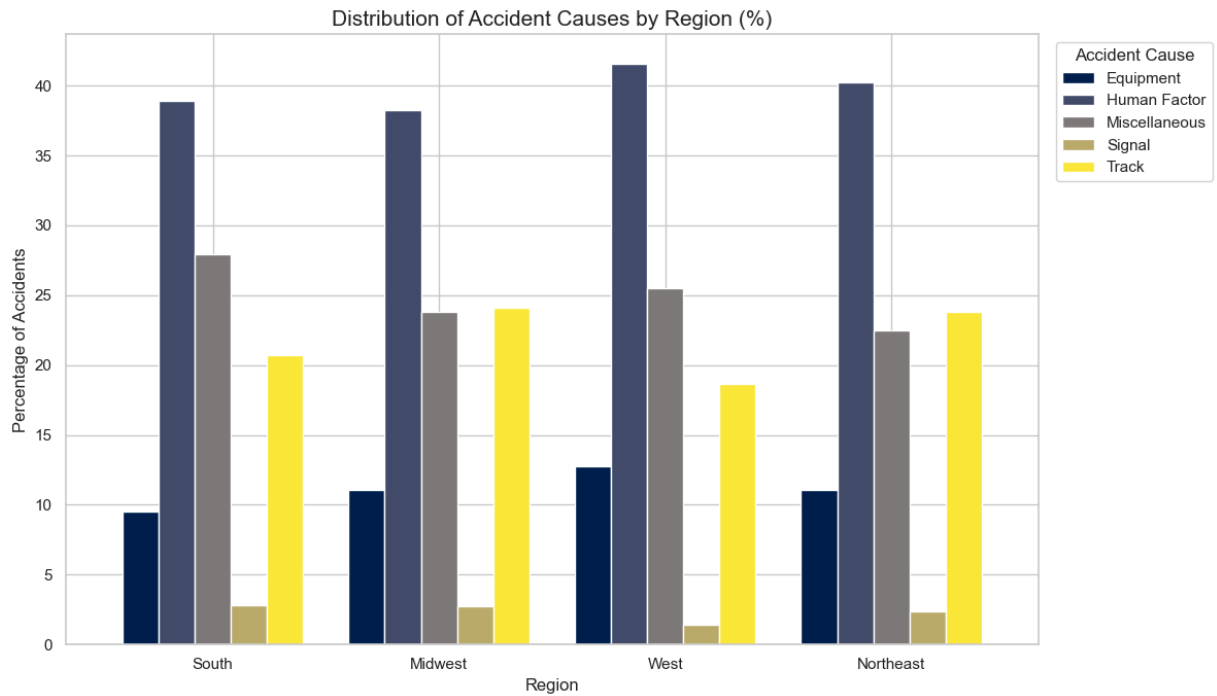


```
In [139... # Calculate the percentage of each cause within each region
# normalize='index' ensures that each row (Region) sums to 1 (100%)
cross_tab = pd.crosstab(df_merged['Region'], df_merged['Cause_Name'], normal

# Reorder the regions to match the previous chart (by total accident count)
# This keeps the visualization consistent
region_order = df_merged['Region'].value_counts().index
cross_tab = cross_tab.loc[region_order]

# Plot as a grouped bar chart
ax = cross_tab.plot(kind='bar', figsize=(12, 7), width=0.8, colormap='civid

# formatting
plt.title('Distribution of Accident Causes by Region (%)', fontsize=15)
plt.ylabel('Percentage of Accidents', fontsize=12)
plt.xlabel('Region', fontsize=12)
plt.legend(title='Accident Cause', bbox_to_anchor=(1.01, 1), loc='upper left
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



Label Encoding Categorical Features

Correlation matrices require numerical inputs, so we convert key categorical features into numbers using Label Encoding. Each unique category gets assigned an integer value.

Note: `nan` values are getting their own encoded value in each column. This will need to be addressed later in preprocessing.

```
In [140... from sklearn.preprocessing import LabelEncoder

# Columns to encode
cols_to_encode = ['Weather Condition', 'Track Type', 'Visibility', 'Accident

# Create a copy so we don't mess with the original
df_encoded = df_merged.copy()

# Label encode each column
le = LabelEncoder()
for col in cols_to_encode:
    df_encoded[col + '_encoded'] = le.fit_transform(df_merged[col].astype(st
    print(f"{col}: {dict(zip(le.classes_, le.transform(le.classes_)))}")
```

```

Weather Condition: {'Clear': np.int64(0), 'Cloudy': np.int64(1), 'Fog': np.int64(2), 'Rain': np.int64(3), 'Sleet': np.int64(4), 'Snow': np.int64(5), nan: np.int64(6)}
Track Type: {'Industry': np.int64(0), 'Main': np.int64(1), 'Siding': np.int64(2), 'Yard': np.int64(3), nan: np.int64(4)}
Visibility: {'Dark': np.int64(0), 'Dawn': np.int64(1), 'Day': np.int64(2), 'Dusk': np.int64(3), nan: np.int64(4)}
Accident Type: {'Broken train collision': np.int64(0), 'Derailment': np.int64(1), 'Explosion-detonation': np.int64(2), 'Fire/violent rupture': np.int64(3), 'Head on collision': np.int64(4), 'Hwy-rail crossing': np.int64(5), 'Obstruction': np.int64(6), 'Other (describe in narrative)': np.int64(7), 'Other impacts': np.int64(8), 'RR grade crossing': np.int64(9), 'Raking collision': np.int64(10), 'Rear end collision': np.int64(11), 'Side collision': np.int64(12), nan: np.int64(13)}
Region: {'Midwest': np.int64(0), 'Northeast': np.int64(1), 'South': np.int64(2), 'West': np.int64(3)}

```

Correlation Matrix - Relevant Features

We check how candidate features relate to each other to spot redundancy early. If two features are highly correlated, they're telling the same story and we may only need one.

Key findings:

- **Track Type & Train Speed (-0.38):** Yard tracks run slower than mainlines - partially redundant.
- **Population & RUCC (-0.37):** Expected overlap - may only need one.
- **Everything else:** Near-zero correlation, meaning each feature brings something unique to the classification task.

```

In [171]: # Clean Total Damage Cost - remove commas and convert to number
df_encoded['Total Damage Cost'] = pd.to_numeric(
    df_encoded['Total Damage Cost'].astype(str).str.replace(',', ''),
    errors='coerce'
)

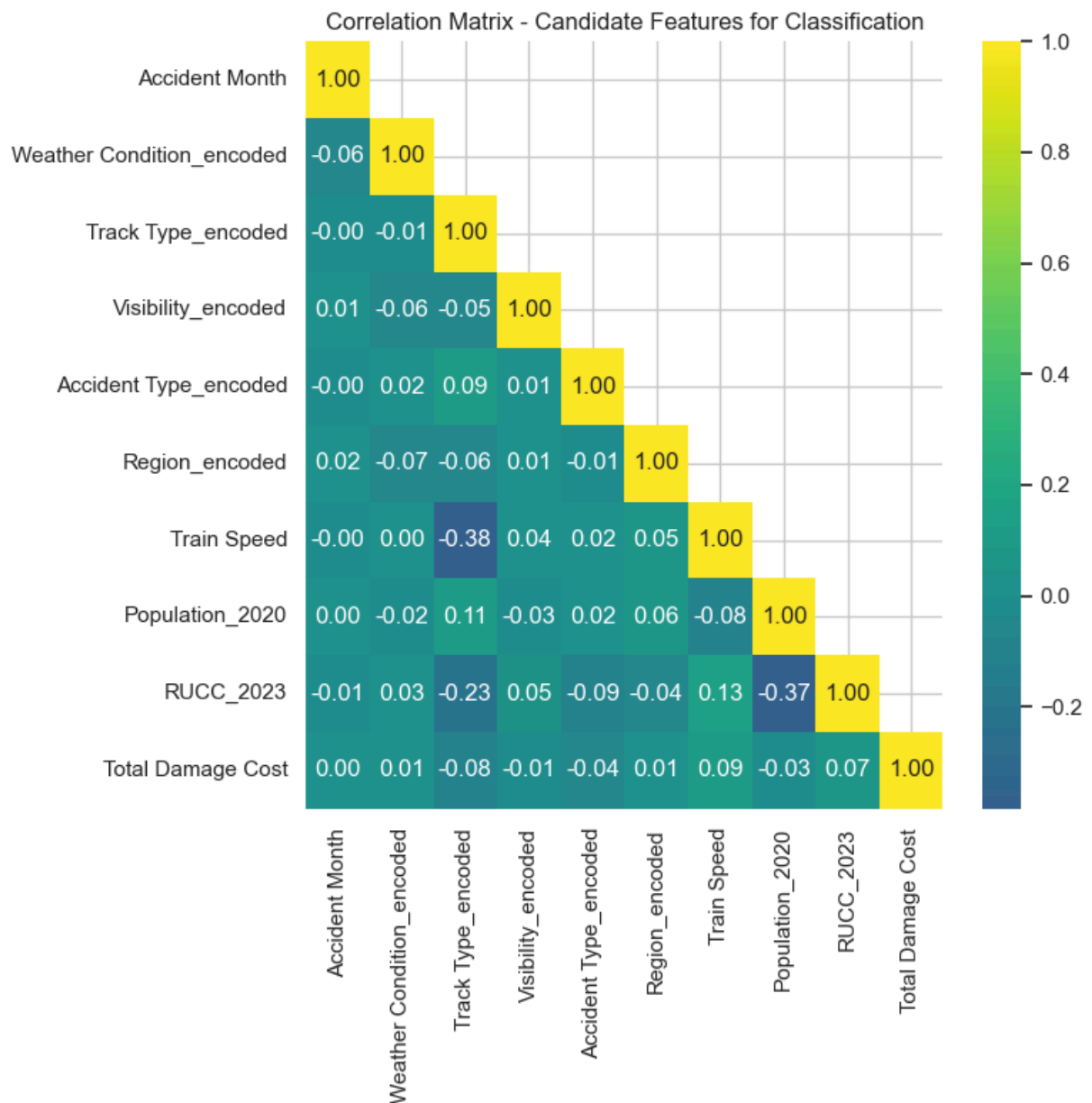
# Narrow down to features relevant to cause code prediction
relevant_cols = [
    'Accident Month',
    'Weather Condition_encoded',
    'Track Type_encoded',
    'Visibility_encoded',
    'Accident Type_encoded',
    'Region_encoded',
    'Train Speed',
    'Population_2020',
    'RUCC_2023',
    'Total Damage Cost'
]

# Create focused correlation matrix
corr_focused = df_encoded[relevant_cols].corr()

```

```
mask = np.triu(np.ones_like(corr_focused, dtype=bool), k=1)

plt.figure(figsize=(8, 8))
sns.heatmap(corr_focused, annot=True, fmt='.2f', cmap='viridis', center=0, n
plt.title('Correlation Matrix - Candidate Features for Classification')
plt.tight_layout()
plt.show()
```



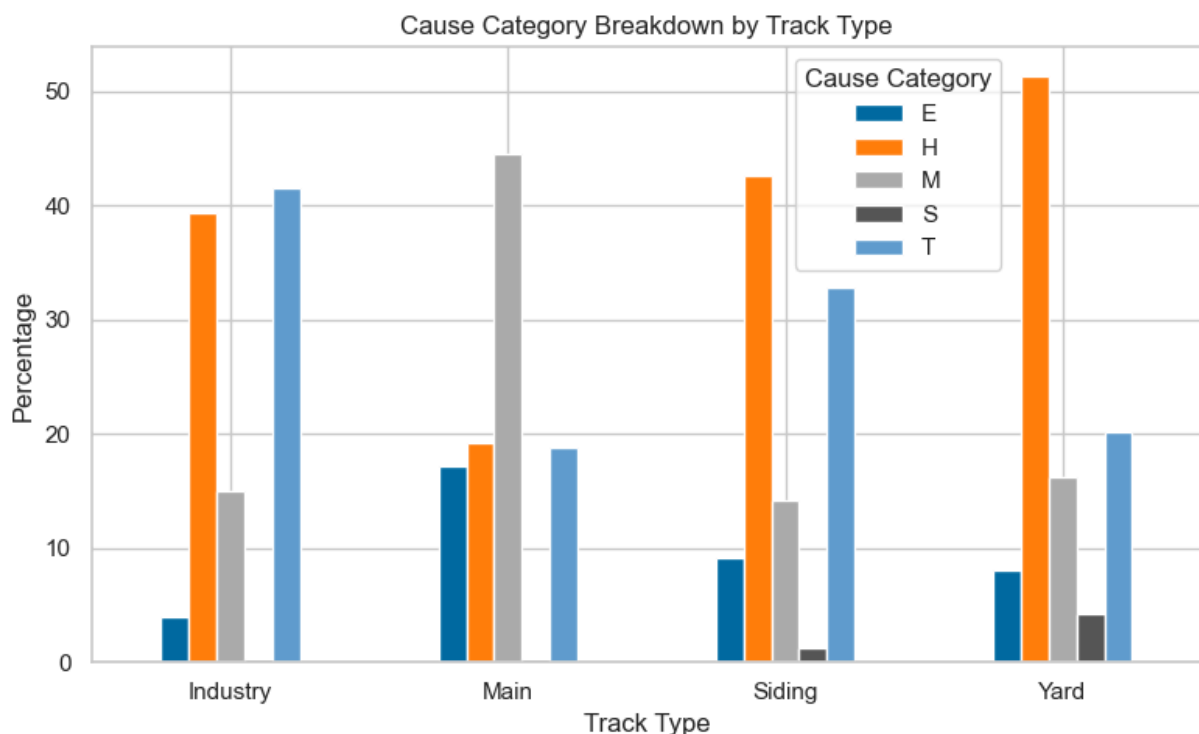
Track Type vs Cause Category

Examining how accident cause categories distribute across different track types. Each bar group adds to 100%, so we're comparing proportions not raw counts.

Key takeaway: Mainline tracks are dominated by Mechanical failures, while Yard and Industry tracks show higher Human Factor rates. This makes operational sense -

mainlines run high-speed freight while yards rely heavily on manual switching movements.

```
In [142... # Track Type vs Cause Category
plt.figure(figsize=(8, 5))
ct = pd.crosstab(df_merged['Track Type'], df_merged['Cause_Category'], norma
ct.plot(kind='bar', ax=plt.gca())
plt.title('Cause Category Breakdown by Track Type')
plt.xlabel('Track Type')
plt.ylabel('Percentage')
plt.xticks(rotation=0)
plt.legend(title='Cause Category', loc='upper right', bbox_to_anchor=(.83,
plt.tight_layout()
plt.show()
```



RUCC Code vs Cause Category

Looking at how cause categories shift across the metro-to-rural spectrum. This builds on Phase 1's statistical finding that metro and rural counties have fundamentally different accident patterns.

Key takeaway: Metro counties (RUCC 1-3) are dominated by Human Factor causes, while rural counties (RUCC 7-9) show a clear increase in Track and Mechanical failures. The middle RUCC codes (4-6) act as a transition zone where causes are more evenly distributed.

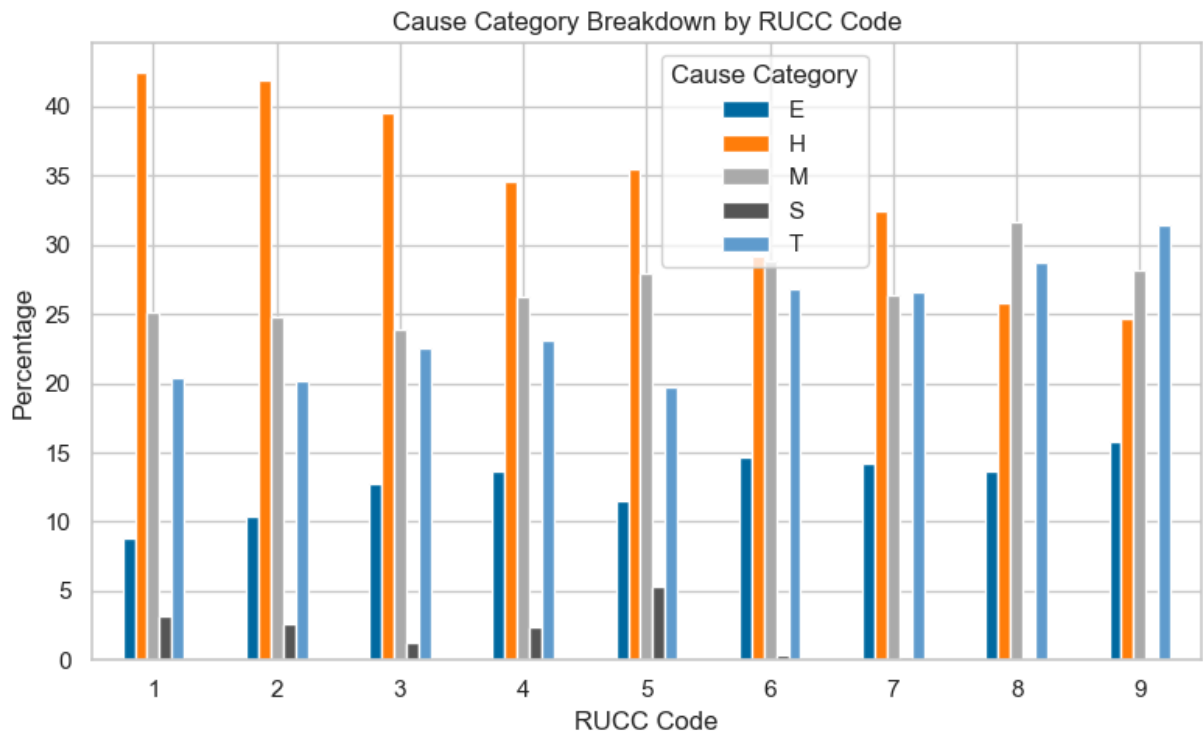
```
In [143... # RUCC vs Cause Category
plt.figure(figsize=(8, 5))
ct_rucc = pd.crosstab(df_merged['RUCC_2023'], df_merged['Cause_Category'], r
```



```

ct_rucc.plot(kind='bar', ax=plt.gca())
plt.title('Cause Category Breakdown by RUCC Code')
plt.xlabel('RUCC Code')
plt.ylabel('Percentage')
plt.xticks(rotation=0)
plt.legend(title='Cause Category', loc='upper right', bbox_to_anchor=(.71,
plt.tight_layout()
plt.show()

```



Feature Engineering: RUCC Metro-Adjacency Classification

The bar chart above shows cause category shifts across all 9 RUCC codes, but for modeling purposes we need to reduce this granularity. Rather than using arbitrary equal-width bins, we follow the USDA's own classification methodology.

The RUCC codes encode two distinct dimensions for nonmetro counties: urban population size AND adjacency to metropolitan areas. Adjacent counties (codes 4, 6, 8) share economic ties with metro areas through labor-force commuting (at least 2% of employed workers commute to central metro counties). Nonadjacent counties (codes 5, 7, 9) lack these ties.

This gives us three meaningful groups:

- **Metro** (codes 1-3): Counties within OMB-defined metropolitan statistical areas, subdivided by metro area population (1M+, 250K-1M, under 250K)
- **Nonmetro Adjacent** (codes 4, 6, 8): Nonmetro counties that physically adjoin a metro area with commuting ties

- **Nonmetro Nonadjacent** (codes 5, 7, 9): Nonmetro counties with no meaningful metro economic connection

This binning preserves the adjacency distinction the USDA deliberately built into the coding system, rather than collapsing it with equal-width bins that would mix adjacent and nonadjacent counties together.

Source: USDA ERS, Rural-Urban Continuum Codes 2023 - Methods documentation.

```
In [144... # Map RUCC codes to Metro-Adjacency classification
# Based on USDA ERS methodology: metro vs nonmetro adjacent vs nonadjacent
rucc_adjacency_map = {
    1: 'Metro',
    2: 'Metro',
    3: 'Metro',
    4: 'Nonmetro_Adjacent',
    5: 'Nonmetro_Nonadjacent',
    6: 'Nonmetro_Adjacent',
    7: 'Nonmetro_Nonadjacent',
    8: 'Nonmetro_Adjacent',
    9: 'Nonmetro_Nonadjacent'
}

df_merged['RUCC_Metro_Adjacency'] = df_merged['RUCC_2023'].map(rucc_adjacency_map)

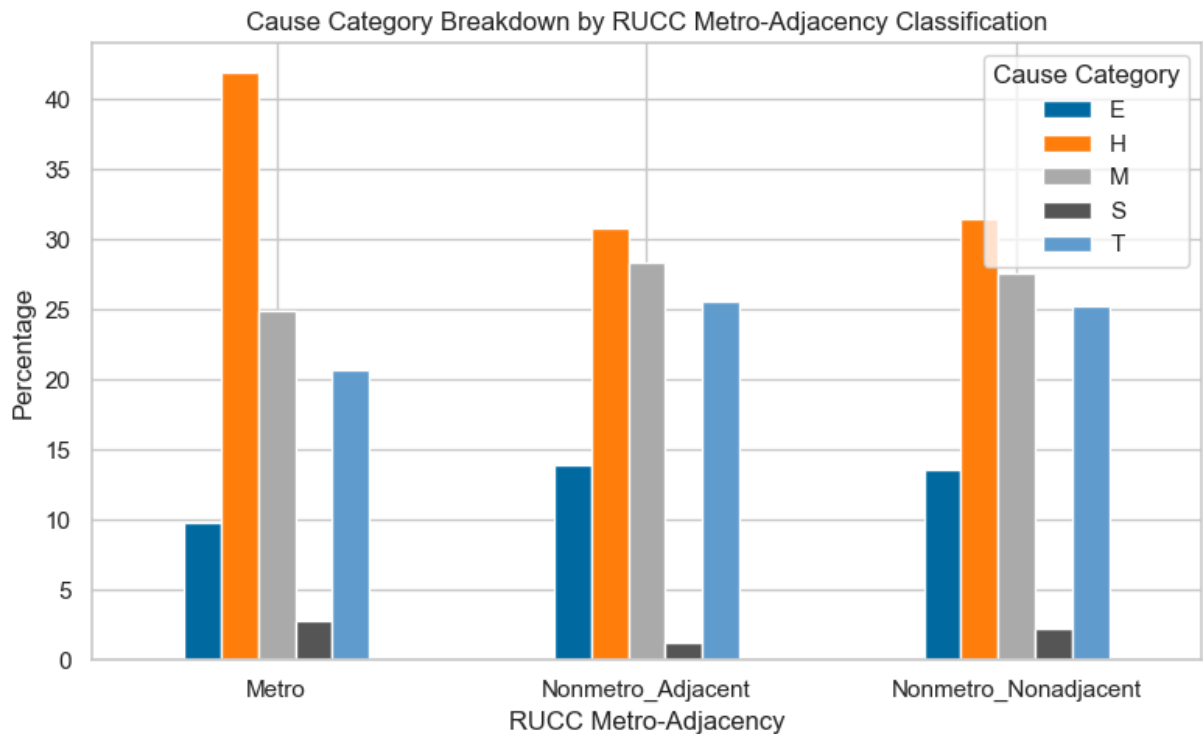
# Quick sanity check - make sure nothing slipped through
print("RUCC_Metro_Adjacency value counts:\n")
print(df_merged['RUCC_Metro_Adjacency'].value_counts())
print(f"\nNull values: {df_merged['RUCC_Metro_Adjacency'].isna().sum()}")
```

RUCC_Metro_Adjacency value counts:

```
RUCC_Metro_Adjacency
Metro                26193
Nonmetro_Adjacent    4505
Nonmetro_Nonadjacent 3864
Name: count, dtype: int64
```

Null values: 0

```
In [145... # RUCC Metro-Adjacency vs Cause Category
plt.figure(figsize=(8, 5))
ct_adjacency = pd.crosstab(df_merged['RUCC_Metro_Adjacency'], df_merged['Cause Category'])
ct_adjacency.plot(kind='bar', ax=plt.gca())
plt.title('Cause Category Breakdown by RUCC Metro-Adjacency Classification')
plt.xlabel('RUCC Metro-Adjacency')
plt.ylabel('Percentage')
plt.xticks(rotation=0)
plt.legend(title='Cause Category', loc='upper right')
plt.tight_layout()
plt.show()
```



Findings: RUCC Metro-Adjacency Feature Engineering

The binned RUCC Metro-Adjacency classification reveals a clear separation between metro and nonmetro accident cause patterns. Metro counties show a notably higher proportion of Human Factor (H) causes (~42%) compared to both nonmetro groups (~31%), while Environmental (E), Mechanical (M), and Track (T) causes all increase in nonmetro counties.

However, the two nonmetro subgroups - Adjacent and Nonadjacent - show nearly identical cause category distributions. This suggests that for accident cause prediction, the primary distinction is metro vs nonmetro rather than adjacency to metro areas. The commuting-based economic ties captured by the adjacency dimension do not appear to meaningfully influence what causes railroad accidents.

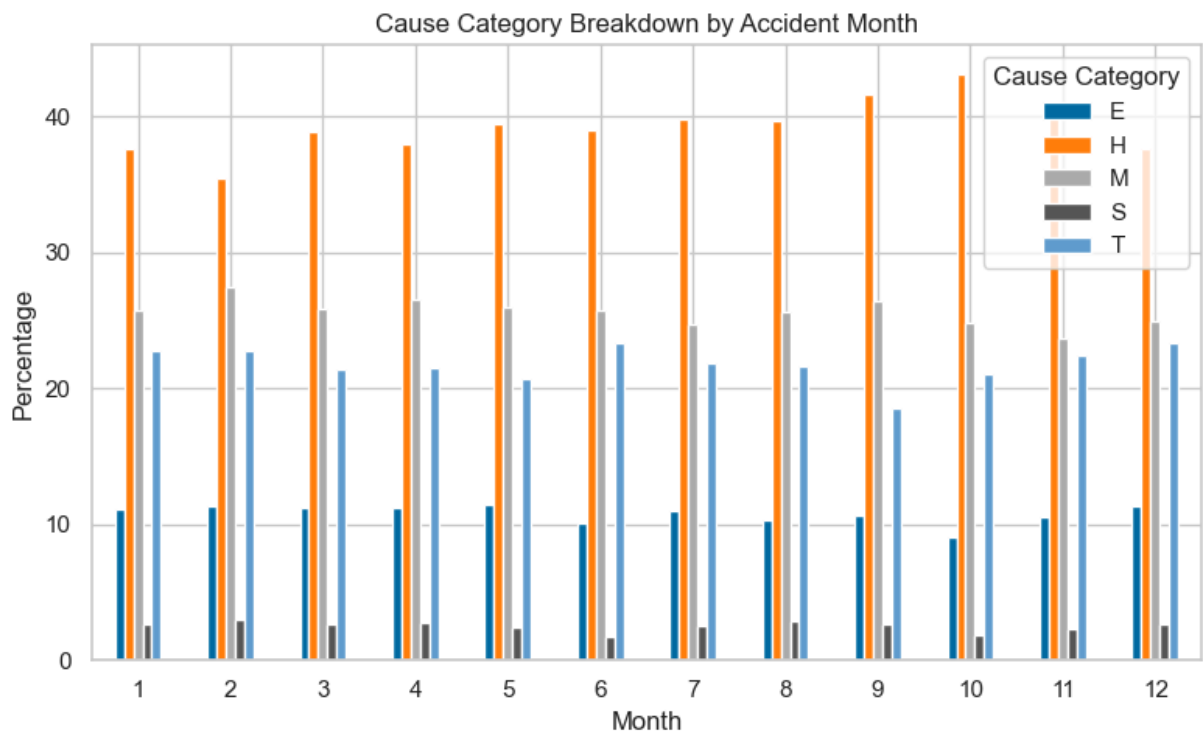
This is a useful finding for modeling. While we retain the three-group classification to stay consistent with USDA methodology, models may effectively treat this as a two-class feature (metro vs nonmetro). If a model collapses the two nonmetro groups, that would be consistent with what the data shows here.

Accident Month vs Cause Category

Checking for seasonal patterns in cause code distribution. The hypothesis was that certain causes (Track failures, Environmental) might shift with seasons due to weather and infrastructure stress.

Key takeaway: Cause category proportions remain relatively stable across all 12 months. Accident Month does not appear to be a strong predictor of cause codes. This is useful to know - it helps us focus on the features that actually matter (Track Type, RUCC) going forward.

```
In [146... # Accident Month vs Cause Category
plt.figure(figsize=(8, 5))
ct_month = pd.crosstab(df_merged['Accident Month'], df_merged['Cause_Categor
ct_month.plot(kind='bar', ax=plt.gca())
plt.title('Cause Category Breakdown by Accident Month')
plt.xlabel('Month')
plt.ylabel('Percentage')
plt.xticks(rotation=0)
plt.legend(title='Cause Category')
plt.tight_layout()
plt.show()
```



Feature Engineering: Seasonal Binning from Months

To capture the impact of annual weather cycles on rail accident causes (specifically Environmental and Track failures), the discrete month of each incident was binned into one of four seasonal categories. This binning was not subjective; it strictly adheres to the National Oceanic and Atmospheric Administration (NOAA) Meteorological Seasons standard (Spring: Mar-May, Summer: Jun-Aug, Autumn: Sep-Nov, Winter: Dec-Feb).

We selected the meteorological standard over the astronomical standard for two methodological reasons:

Climatological Alignment: NOAA uses meteorological seasons specifically because they naturally align with annual temperature cycles and weather extremes. Since our goal is to capture the environmental strain on infrastructure at the time of the accident (e.g., thermal track buckling in peak summer heat, or switch freezing in deep winter), aligning our bins with the standard used by climatologists provides the most scientifically valid feature representation.

Calendar Boundary Consistency: Astronomical seasons change late in the month (e.g., December 21). If we used astronomical seasons, an accident occurring on December 15 during a severe snowstorm would be classified as an Autumn event. The meteorological standard ensures that complete calendar months with consistent weather profiles are grouped intact, preventing temporal misclassification of the incident's environment.

Source: NOAA/NCEI: Meteorological Versus Astronomical Seasons

```
In [147... # Map Accident Month to NOAA Meteorological Seasons
# Winter: Dec(12), Jan(1), Feb(2)
# Spring: Mar(3), Apr(4), May(5)
# Summer: Jun(6), Jul(7), Aug(8)
# Autumn: Sep(9), Oct(10), Nov(11)
season_map = {
    12: 'Winter', 1: 'Winter', 2: 'Winter',
    3: 'Spring', 4: 'Spring', 5: 'Spring',
    6: 'Summer', 7: 'Summer', 8: 'Summer',
    9: 'Autumn', 10: 'Autumn', 11: 'Autumn'
}

df_merged['Season'] = df_merged['Accident Month'].map(season_map)

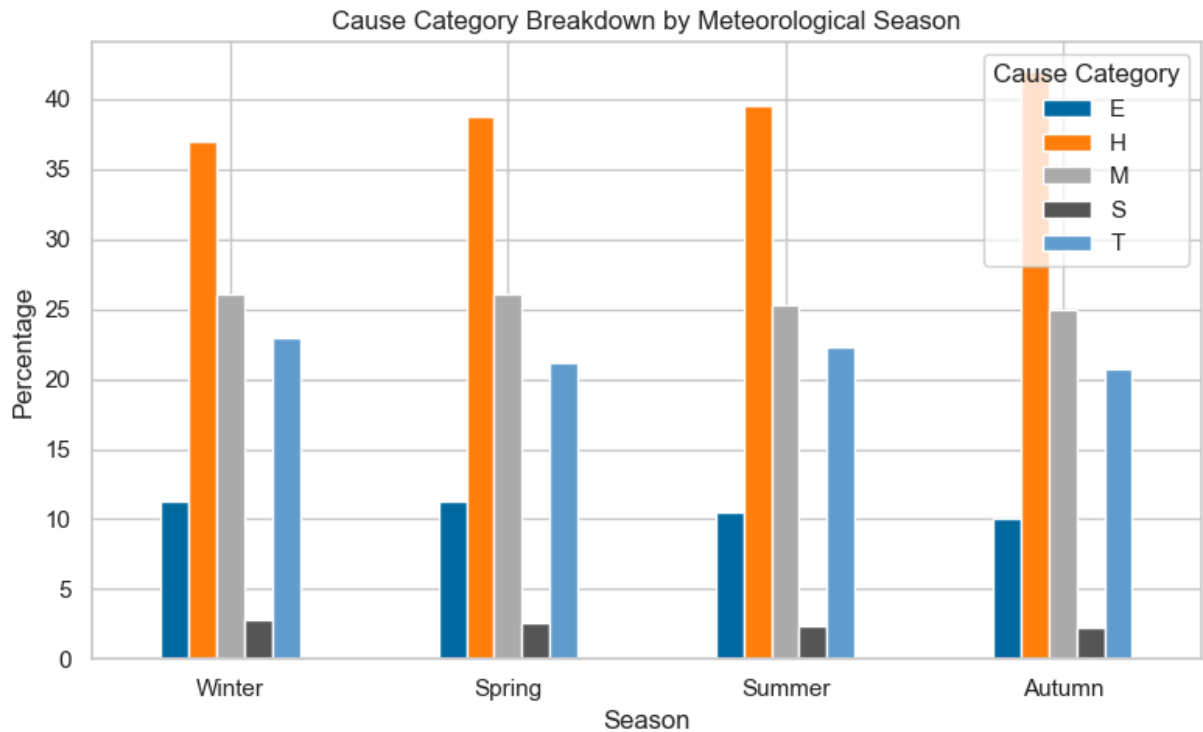
# Sanity check
print("Season value counts:")
print(df_merged['Season'].value_counts())
print(f"\nNull values: {df_merged['Season'].isna().sum()}")
```

```
Season value counts:
Season
Summer    9306
Spring    8554
Winter     8435
Autumn     8267
Name: count, dtype: int64
```

```
Null values: 0
```

```
In [148... # Season vs Cause Category
plt.figure(figsize=(8, 5))
ct_season = pd.crosstab(df_merged['Season'], df_merged['Cause_Category'], normalize=True)
ct_season = ct_season.reindex(['Winter', 'Spring', 'Summer', 'Autumn'])
ct_season.plot(kind='bar', ax=plt.gca())
plt.title('Cause Category Breakdown by Meteorological Season')
plt.xlabel('Season')
plt.ylabel('Percentage')
plt.xticks(rotation=0)
```

```
plt.legend(title='Cause Category', loc='upper right')
plt.tight_layout()
plt.show()
```



Findings: Seasonal Binning Feature Engineering

The seasonal breakdown confirms what the raw monthly analysis suggested - cause category proportions are remarkably stable across all four meteorological seasons. Human Factor (H) remains the dominant cause in every season (~37-42%), followed by Mechanical (M), Track (T), Environmental (E), and Signal (S) in consistent order.

Despite the climatological rationale for expecting seasonal variation - thermal track buckling in summer, switch freezing in winter - the data does not show meaningful shifts in cause category distribution by season. This suggests that while weather may influence accident frequency or severity, it does not strongly determine the type of cause.

Season will be retained as a candidate feature for modeling, but based on this visualization we should not expect it to be a strong predictor of cause category. If models assign it low importance, this chart explains why.

Feature Engineering: Train Composition Metrics

Created three new features to capture train loading characteristics, which may relate to accident risk through factors like slack action, train handling difficulty, and weight distribution:

- **total_cars**: Sum of all cars in the consist (loaded freight, empty freight, loaded passenger, empty passenger, and cabooses)
- **empty_loaded_ratio**: Ratio of empty cars to loaded cars (added 1 to denominator to avoid division by zero)
- **pct_empty**: Percentage of cars in the consist that are empty

These features allow us to explore whether train composition (particularly the proportion of empty cars) differs across accident cause categories.

```
In [149... # Total cars in consist
df_merged['total_cars'] = (df_merged['Loaded Freight Cars'] + df_merged['Empty Freight Cars'] +
                           df_merged['Loaded Passenger Cars'] + df_merged['Empty Passenger Cars'] +
                           df_merged['Cabooses'])

# Empty to loaded ratio
df_merged['empty_loaded_ratio'] = (df_merged['Empty Freight Cars'] + df_merged['Empty Passenger Cars'] +
                                   df_merged['Cabooses']) / (df_merged['Loaded Freight Cars'] + df_merged['Loaded Passenger Cars'] + df_merged['Cabooses'])

# Or percentage empty
df_merged['pct_empty'] = ((df_merged['Empty Freight Cars'] + df_merged['Empty Passenger Cars'] + df_merged['Cabooses']) / df_merged['total_cars']) * 100
```

Distribution of Train Composition Features

Summary statistics for the newly created train composition features:

```
In [150... # Quick look at the distributions - side by side
summary_stats = pd.DataFrame({
    'total_cars': df_merged['total_cars'].describe(),
    'empty_loaded_ratio': df_merged['empty_loaded_ratio'].describe(),
    'pct_empty': df_merged['pct_empty'].describe()
})

print(summary_stats)
print(f"\nZero cars: {(df_merged['total_cars'] == 0).sum()}")
print(f"Infinite values in empty_loaded_ratio: {np.isinf(df_merged['empty_loaded_ratio']).sum()}")
```

	total_cars	empty_loaded_ratio	pct_empty
count	34562.000000	34562.000000	30430.000000
mean	43.22898	4.882367	39.055297
std	48.34557	19.945770	38.797730
min	0.000000	0.000000	0.000000
25%	2.000000	0.000000	0.000000
50%	24.000000	0.214286	30.000000
75%	76.000000	1.071429	74.358974
max	832.000000	832.000000	100.000000

Zero cars: 4132

Infinite values in empty_loaded_ratio: 0

Train Composition Features by Cause Category

```
In [151... # Group by your cause categories
print("\n=== Empty/Loaded Ratio by Cause Category ===")
print(df_merged.groupby('Cause_Category')['empty_loaded_ratio'].agg(['mean',

print("\n=== Percent Empty by Cause Category ===")
print(df_merged.groupby('Cause_Category')['pct_empty'].agg(['mean', 'median']
```

```
=== Empty/Loaded Ratio by Cause Category ===
```

	mean	median	count
Cause_Category			
E	4.781247	0.285714	3721
H	4.732267	0.433333	13582
M	6.531271	0.011364	8856
S	1.651375	0.470588	867
T	3.637779	0.039268	7534

```
=== Percent Empty by Cause Category ===
```

	mean	median	count
Cause_Category			
E	36.612395	29.357798	3399
H	46.602091	42.635005	11776
M	36.656126	20.000000	7730
S	42.755492	37.735849	829
T	29.343408	13.513514	6694

Statistical Significance Test: Percent Empty Across Cause Categories

ANOVA test using `scipy.stats` to determine if percent empty differs significantly across cause categories.

```
In [152... from scipy.stats import f_oneway

# Get the empty_loaded_ratio for each cause category
groups = [group['pct_empty'].dropna()
           for name, group in df_merged.groupby('Cause_Category')]

f_stat, p_value = f_oneway(*groups)
print(f"\nANOVA F-statistic: {f_stat:.4f}")
print(f"P-value: {p_value:.6f}")

if p_value < 0.05:
    print("Percent empty feature differs significantly across cause categories")
else:
    print("No significant difference")
```

```
ANOVA F-statistic: 235.9680
```

```
P-value: 0.000000
```

```
Percent empty feature differs significantly across cause categories
```

Data Cleaning: Gross Tonnage Conversion

Convert Gross Tonnage from string format (with commas) to numeric, then calculate tonnage per car as an additional train characteristic feature.

```
In [153... # # Convert Gross Tonnage to numeric (removes commas)
# df_merged['Gross Tonnage'] = pd.to_numeric(df_merged['Gross Tonnage'].str.

# # Now create tonnage per car
# df_merged['tonnage_per_car'] = df_merged['Gross Tonnage'] / (df_merged['to

# # Check it worked
# print(df_merged['Gross Tonnage'].dtype)
# print(df_merged['Gross Tonnage'].head())

# Convert Gross Tonnage to numeric (handle mixed types by converting to stri
df_merged['Gross Tonnage'] = pd.to_numeric(
    df_merged['Gross Tonnage'].astype(str).str.replace(',', ''),
    errors='coerce'
)

# Now create tonnage per car
df_merged['tonnage_per_car'] = df_merged['Gross Tonnage'] / (df_merged['tota

# Check it worked
print(df_merged['Gross Tonnage'].dtype)
print(df_merged['Gross Tonnage'].head())
```

int64

```
0    1665
1    8691
2         0
3         0
4         0
```

Name: Gross Tonnage, dtype: int64

Correlation Analysis: Train Composition Features

Examine correlations between newly created train composition features and existing features to assess redundancy and relationships.

```
In [154... # Create tonnage per car feature
df_merged['tonnage_per_car'] = df_merged['Gross Tonnage'] / (df_merged['tota

# Correlation matrix for your new features
features = ['empty_loaded_ratio', 'pct_empty', 'total_cars',
            'Gross Tonnage', 'tonnage_per_car', 'Train Speed']

corr_matrix = df_merged[features].corr()
print("\n=== Correlation Matrix ===")
print(corr_matrix)
```

=== Correlation Matrix ===

	empty_loaded_ratio	pct_empty	total_cars	Gross Tonnage
\				
empty_loaded_ratio	1.000000	0.388616	0.264686	-0.021009
pct_empty	0.388616	1.000000	-0.044848	-0.250400
total_cars	0.264686	-0.044848	1.000000	0.759903
Gross Tonnage	-0.021009	-0.250400	0.759903	1.000000
tonnage_per_car	-0.027470	-0.084686	0.051630	0.302444
Train Speed	0.038338	-0.204342	0.150901	0.155593

	tonnage_per_car	Train Speed
empty_loaded_ratio	-0.027470	0.038338
pct_empty	-0.084686	-0.204342
total_cars	0.051630	0.150901
Gross Tonnage	0.302444	0.155593
tonnage_per_car	1.000000	0.019575
Train Speed	0.019575	1.000000

Feature Correlation Heatmap: Candidate Features for Classification

Transfer new train composition features to the encoded dataframe and visualize correlations with existing candidate features to assess which features provide unique predictive information for cause code classification.

```
In [168... # First, merge the new features into df_encoded
# (assuming df_encoded and df_merged share the same index/can be aligned)

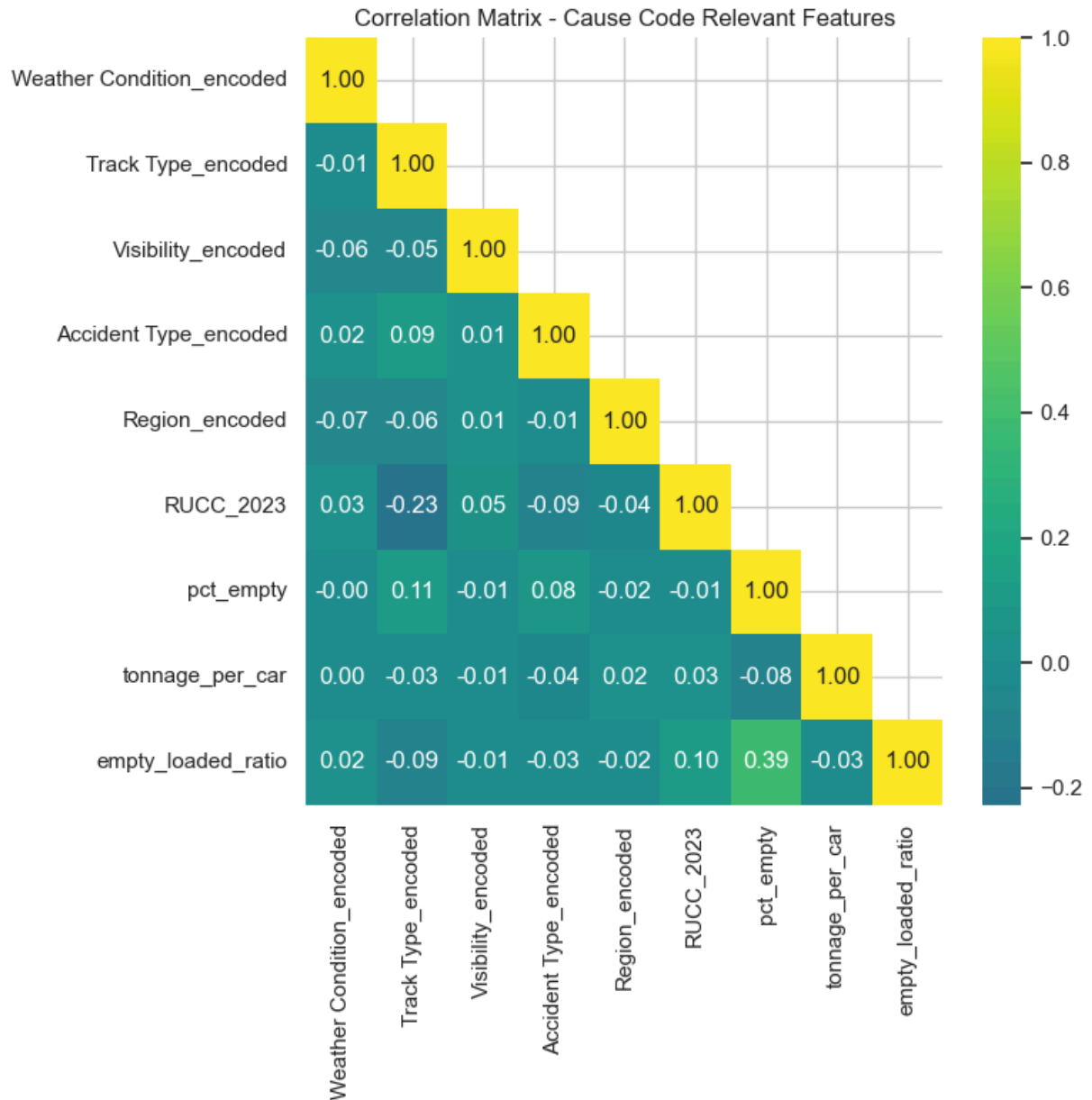
df_encoded['pct_empty'] = df_merged['pct_empty']
df_encoded['tonnage_per_car'] = df_merged['tonnage_per_car']
df_encoded['empty_loaded_ratio'] = df_merged['empty_loaded_ratio']

# Now add them to relevant_cols
relevant_cols = [
    'Weather Condition_encoded',
    'Track Type_encoded',
    'Visibility_encoded',
    'Accident Type_encoded',
    'Region_encoded',
    'RUCC_2023',
    'pct_empty',
    'tonnage_per_car',
    'empty_loaded_ratio'
]

# Create correlation matrix
corr_focused = df_encoded[relevant_cols].corr()
mask = np.triu(np.ones_like(corr_focused, dtype=bool), k=1)

plt.figure(figsize=(8, 8))
sns.heatmap(corr_focused, annot=True, fmt='.2f', cmap='viridis', center=0, n
plt.title('Correlation Matrix - Cause Code Relevant Features')
```

```
plt.tight_layout()
plt.show()
```



Summary: Train Composition Feature Selection

We explored three train composition features to capture loading characteristics that may relate to accident causes through factors like slack action and train handling difficulty:

1. **total_cars**: Total consist length
2. **empty_loaded_ratio**: Ratio of empty to loaded cars
3. **pct_empty**: Percentage of cars that are empty

Analysis Steps:

Initial distribution analysis revealed substantial variation in train composition, with a median train consisting of 24 cars and 30% empty. Approximately 12% of accidents

involved zero cars, likely yard incidents or single locomotive events.

Grouping by cause category showed meaningful patterns: Track (T) and Mechanical (M) accidents occurred with heavier trains (median 13.5% and 20% empty), while Human Factor (H) accidents occurred with lighter consists (median 42.6% empty). ANOVA testing confirmed these differences were statistically significant (F-statistic: 235.9, $p < 0.001$).

Correlation analysis between the three candidate features and existing encoded features revealed that **pct_empty** was the strongest candidate because:

- Showed moderate correlation with Track Type (0.11) and RUCC codes (0.10), providing complementary information to existing features
- More interpretable than empty_loaded_ratio
- tonnage_per_car showed essentially no correlation with any features

Based on these findings, **pct_empty** would be our choice as the train composition feature to carry forward for potential inclusion in the classification model. Although, the inclusion of this feature could bias our association rules towards freight-focused accidents.

Feature Engineering: Temporal and Operational Features

Building on the initial feature exploration, we now engineer additional categorical features that capture temporal patterns and operational characteristics. This section creates and evaluates candidate features for inclusion in the classification model:

Features to engineer:

- **Equipment Type:** Categorize train operations (freight, passenger, yard switching, maintenance)
- **Time of Day:** Bin accident times into four periods (Morning, Afternoon, Evening, Night)
- **Day of Week:** Extract weekday patterns from accident dates
- **Visibility:** Clean and encode existing visibility conditions
- **Signalization:** Clean and encode track signalization status

Analysis approach: After feature creation, we'll encode all features and examine their relationships through correlation analysis. Statistical significance testing (chi-square with Bonferroni correction) will determine which features show meaningful associations with accident cause categories while accounting for multiple comparisons.

Map Equipment Type codes to descriptive labels, and verify **Visibility** and **Signalization** are clean categorical features ready for encoding.

```
In [156... # Map Equipment Type codes to labels (fix keys to strings)
equipment_type_map = {
    '1': 'Freight Train',
    '2': 'Passenger Train-Pulling',
    '3': 'Commuter Train-Pulling',
    '4': 'Work Train',
    '5': 'Single Car',
    '6': 'Cut of Cars',
    '7': 'Yard/Switching',
    '8': 'Light Loco(s)',
    '9': 'Maint./Inspect. Car',
    'A': 'Spec. MoW Equip.',
    'B': 'Passenger Train-Pushing',
    'C': 'Commuter Train-Pushing',
    'D': 'EMU',
    'E': 'DMU'
}

df_merged['Equipment_Type_Mapped'] = df_merged['Equipment Type Code'].map(ec

print("Equipment Type Mapped counts:")
print(df_merged['Equipment_Type_Mapped'].value_counts())
print(f"\nUnmapped (NaN): {df_merged['Equipment_Type_Mapped'].isna().sum()}")
```

```
Equipment Type Mapped counts:
Equipment_Type_Mapped
Freight Train          13699
Yard/Switching         9516
Cut of Cars            2141
Single Car             1931
Light Loco(s)          1695
Passenger Train-Pulling 1103
Spec. MoW Equip.        652
EMU                     441
Maint./Inspect. Car     267
Commuter Train-Pulling  236
Commuter Train-Pushing  207
Work Train             195
Passenger Train-Pushing 191
DMU                     52
Name: count, dtype: int64
```

```
Unmapped (NaN): 2236
```

```
In [157... # Check Visibility output
print("\nVisibility values:")
print(df_merged['Visibility'].value_counts())

# Check Signalization output
print("\nSignalization values:")
print(df_merged['Signalization'].value_counts())
```

```
Visibility values:
Visibility
Day      15515
Dark     12227
Dawn      3512
Dusk      3300
Name: count, dtype: int64
```

```
Signalization values:
Signalization
Not Signaled    24322
Signaled        10238
Name: count, dtype: int64
```

Extract **day of week** from the Date field to capture potential weekly patterns in accident causes (e.g., weekday vs. weekend operations, maintenance schedules).

```
In [158... # Extract day of week from Date field
df_merged['Date'] = pd.to_datetime(df_merged['Date'])
df_merged['Day_of_Week'] = df_merged['Date'].dt.day_name()

print("Day of Week distribution:")
print(df_merged['Day_of_Week'].value_counts())
```

```
Day of Week distribution:
Day_of_Week
Friday      5239
Tuesday     5175
Thursday    5168
Wednesday   5154
Monday      5108
Saturday    4550
Sunday      4168
Name: count, dtype: int64
```

Extract **hour of day** from Time field and **bin into four daily periods: Morning (4am-10am), Afternoon (10am-4pm), Evening (4pm-10pm), and Night (10pm-4am)** to capture temporal patterns in accident causes.

Save all newly engineered features to the merged dataframe for subsequent encoding and analysis.

```
In [159... # Save updated df_merged with all new features
df_merged.to_csv("../data/processed/df_merged_features_complete.csv.zip", ir
print(f"Saved df_merged with all features: {len(df_merged.columns)} columns"

print("\nSaved df_merged with new features:")
print("- Equipment_Type_Mapped")
print("- Day_of_Week")
print("- total_cars")
print("- empty_loaded_ratio")
print("- pct_empty")
print("- tonnage_per_car")
print("- RUCC_Metro_Adjacency")
```

```
print("- Season")
print(f"\nTotal columns: {len(df_merged.columns)}")
```

Saved df_merged with all features: 175 columns

Saved df_merged with new features:

- Equipment_Type_Mapped
- Day_of_Week
- total_cars
- empty_loaded_ratio
- pct_empty
- tonnage_per_car
- RUCC_Metro_Adjacency
- Season

Total columns: 175

Encode newly created categorical features and existing Visibility and Signalization features using LabelEncoder to prepare them for correlation analysis and modeling.

```
In [160... # Encode new categorical features using LabelEncoder

# Initialize encoders
le_equipment = LabelEncoder()
le_day = LabelEncoder()
le_visibility = LabelEncoder()
le_signalization = LabelEncoder()
le_region = LabelEncoder()
le_rucc_adjacency = LabelEncoder()
le_season = LabelEncoder()

# Encode features (handling NaN values)
df_encoded['Equipment_Type_encoded'] = le_equipment.fit_transform(df_merged['Equipment_Type'])
df_encoded['Day_of_Week_encoded'] = le_day.fit_transform(df_merged['Day_of_Week'])
df_encoded['Visibility_encoded'] = le_visibility.fit_transform(df_merged['Visibility'])
df_encoded['Signalization_encoded'] = le_signalization.fit_transform(df_merged['Signalization'])
df_encoded['Region_encoded'] = le_region.fit_transform(df_merged['Region'])
df_encoded['RUCC_Metro_Adjacency_encoded'] = le_rucc_adjacency.fit_transform(df_merged['RUCC_Metro_Adjacency'])
df_encoded['Season_encoded'] = le_season.fit_transform(df_merged['Season'])

df_encoded.to_csv("../data/processed/df_encoded_features_complete.csv.zip",
print(f"Saved df_encoded: {len(df_encoded.columns)} columns\n")

print("Encoded features:")
print(f"Equipment Type: {dict(zip(le_equipment.classes_, le_equipment.transform(df_encoded['Equipment_Type_encoded'])))}")
print(f"Day of Week: {dict(zip(le_day.classes_, le_day.transform(df_encoded['Day_of_Week_encoded'])))}")
print(f"Visibility: {dict(zip(le_visibility.classes_, le_visibility.transform(df_encoded['Visibility_encoded'])))}")
print(f"Signalization: {dict(zip(le_signalization.classes_, le_signalization.transform(df_encoded['Signalization_encoded'])))}")
print(f"Region: {dict(zip(le_region.classes_, le_region.transform(df_encoded['Region_encoded'])))}")
print(f"RUCC Metro-Adjacency: {dict(zip(le_rucc_adjacency.classes_, le_rucc_adjacency.transform(df_encoded['RUCC_Metro_Adjacency_encoded'])))}")
print(f"Season: {dict(zip(le_season.classes_, le_season.transform(df_encoded['Season_encoded'])))}")
```

Saved df_encoded: 183 columns

Encoded features:

Equipment Type: {'Commuter Train-Pulling': np.int64(0), 'Commuter Train-Pushing': np.int64(1), 'Cut of Cars': np.int64(2), 'DMU': np.int64(3), 'EMU': np.int64(4), 'Freight Train': np.int64(5), 'Light Loco(s)': np.int64(6), 'Maint./Inspect. Car': np.int64(7), 'Passenger Train-Pulling': np.int64(8), 'Passenger Train-Pushing': np.int64(9), 'Single Car': np.int64(10), 'Spec. MoW Equip.': np.int64(11), 'Unknown': np.int64(12), 'Work Train': np.int64(13), 'Yard/Switching': np.int64(14)}

Day of Week: {'Friday': np.int64(0), 'Monday': np.int64(1), 'Saturday': np.int64(2), 'Sunday': np.int64(3), 'Thursday': np.int64(4), 'Tuesday': np.int64(5), 'Wednesday': np.int64(6)}

Visibility: {'Dark': np.int64(0), 'Dawn': np.int64(1), 'Day': np.int64(2), 'Dusk': np.int64(3), 'Unknown': np.int64(4)}

Signalization: {'Not Signaled': np.int64(0), 'Signaled': np.int64(1), 'Unknown': np.int64(2)}

Region: {'Midwest': np.int64(0), 'Northeast': np.int64(1), 'South': np.int64(2), 'West': np.int64(3)}

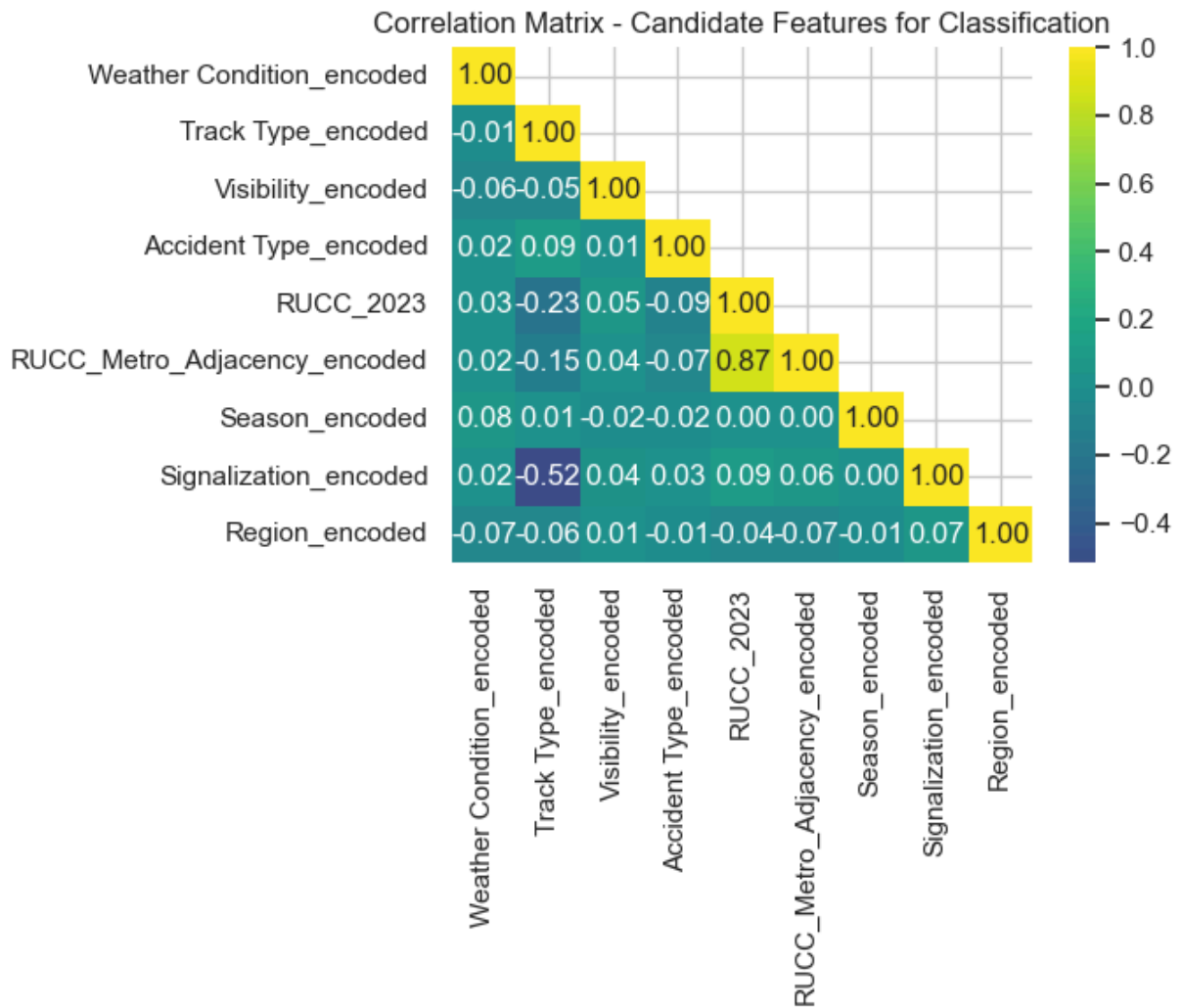
RUCC Metro-Adjacency: {'Metro': np.int64(0), 'Nonmetro_Adjacent': np.int64(1), 'Nonmetro_Nonadjacent': np.int64(2)}

Season: {'Autumn': np.int64(0), 'Spring': np.int64(1), 'Summer': np.int64(2), 'Winter': np.int64(3)}

```
In [172... # Create correlation matrix with all candidate features
relevant_cols = [
    'Weather Condition_encoded',
    'Track Type_encoded',
    'Visibility_encoded',
    'Accident Type_encoded',
    'RUCC_2023',
    'RUCC_Metro_Adjacency_encoded',
    'Season_encoded',
    'Signalization_encoded',
    'Region_encoded'
]

# Create correlation matrix
corr_focused = df_encoded[relevant_cols].corr()
mask = np.triu(np.ones_like(corr_focused, dtype=bool), k=1)

plt.figure(figsize=(7, 6))
sns.heatmap(corr_focused, annot=True, fmt='.2f', cmap='viridis', center=0, n
plt.title('Correlation Matrix - Candidate Features for Classification')
plt.tight_layout()
plt.show()
```

Correlation Significance Testing with Bonferroni Correction

Calculate p-values for all pairwise correlations in the feature correlation matrix to determine which relationships are statistically significant rather than due to chance.

Bonferroni Correction: When conducting multiple statistical tests simultaneously, the probability of obtaining at least one false positive (Type I error) increases with each additional test. The Bonferroni correction addresses this "multiple comparisons problem" by dividing the standard significance threshold ($\alpha = 0.05$) by the number of tests being performed. This creates a more conservative threshold that maintains the overall family-wise error rate at 5%.

For our analysis with 36 pairwise feature comparisons, the Bonferroni-corrected threshold becomes $0.05/36 = 0.00139$. This stricter criterion ensures that correlations we identify as significant are unlikely to be false discoveries, even when testing many relationships simultaneously.

Purpose: We're identifying which feature pairs show genuine statistical relationships that survive the conservative Bonferroni threshold. This helps us understand which features may be redundant (highly correlated with each other) versus which provide independent predictive information. Features with strong, significant correlations are candidates for exclusion to avoid multicollinearity in our classification models.

Interpreting results: P-values below the Bonferroni threshold (0.00139) indicate statistically significant correlations that are highly unlikely to be due to chance. Feature pairs with both high correlation coefficients (e.g., $|r| > 0.5$) and significant p-values represent strong redundancy - these features measure similar underlying information and including both in a model would be problematic. Conversely, features showing no significant correlations with others provide independent predictive information.

```
In [173... # Calculate p-values for each pairwise correlation
n_features = len(relevant_cols)
p_values = np.zeros((n_features, n_features))

for i in range(n_features):
    for j in range(n_features):
        if i != j:
            _, p_val = pearsonr(df_encoded[relevant_cols[i]].dropna(),
                               df_encoded[relevant_cols[j]].dropna())
            p_values[i, j] = p_val
        else:
            p_values[i, j] = 0 # diagonal

# Create DataFrame for p-values
p_values_df = pd.DataFrame(p_values,
                           index=relevant_cols,
                           columns=relevant_cols)

# Apply Bonferroni correction
n_tests = (n_features * (n_features - 1)) / 2 # number of unique pairs
bonferroni_alpha = 0.05 / n_tests

print(f"Number of correlation tests: {n_tests}")
print(f"Bonferroni corrected alpha: {bonferroni_alpha:.6f}")
print("\nP-values for correlations:")
print(p_values_df.round(6))
```

Number of correlation tests: 36.0
Bonferroni corrected alpha: 0.001389

P-values for correlations:

	Weather Condition_encoded	Track Type_encoded
\		
Weather Condition_encoded	0.000000	0.040132
Track Type_encoded	0.040132	0.000000
Visibility_encoded	0.000000	0.000000
Accident Type_encoded	0.003222	0.000000
RUCC_2023	0.000001	0.000000
RUCC_Metro_Adjacency_encoded	0.002731	0.000000
Season_encoded	0.000000	0.325657
Signalization_encoded	0.000006	0.000000
Region_encoded	0.000000	0.000000

	Visibility_encoded	Accident Type_encoded	\
Weather Condition_encoded	0.000000	0.003222	
Track Type_encoded	0.000000	0.000000	
Visibility_encoded	0.000000	0.240852	
Accident Type_encoded	0.240852	0.000000	
RUCC_2023	0.000000	0.000000	
RUCC_Metro_Adjacency_encoded	0.000000	0.000000	
Season_encoded	0.000161	0.000169	
Signalization_encoded	0.000000	0.000000	
Region_encoded	0.065006	0.009652	

	RUCC_2023	RUCC_Metro_Adjacency_encoded	\
Weather Condition_encoded	0.000001	0.002731	
Track Type_encoded	0.000000	0.000000	
Visibility_encoded	0.000000	0.000000	
Accident Type_encoded	0.000000	0.000000	
RUCC_2023	0.000000	0.000000	
RUCC_Metro_Adjacency_encoded	0.000000	0.000000	
Season_encoded	0.834086	0.710264	
Signalization_encoded	0.000000	0.000000	
Region_encoded	0.000000	0.000000	

	Season_encoded	Signalization_encoded	\
Weather Condition_encoded	0.000000	0.000006	
Track Type_encoded	0.325657	0.000000	
Visibility_encoded	0.000161	0.000000	
Accident Type_encoded	0.000169	0.000000	
RUCC_2023	0.834086	0.000000	
RUCC_Metro_Adjacency_encoded	0.710264	0.000000	
Season_encoded	0.000000	0.749515	
Signalization_encoded	0.749515	0.000000	
Region_encoded	0.007206	0.000000	

	Region_encoded
Weather Condition_encoded	0.000000
Track Type_encoded	0.000000
Visibility_encoded	0.065006
Accident Type_encoded	0.009652
RUCC_2023	0.000000
RUCC_Metro_Adjacency_encoded	0.000000

Season_encoded	0.007206
Signalization_encoded	0.000000
Region_encoded	0.000000

Display pairwise correlations alongside their statistical significance after Bonferroni correction to identify which feature relationships are genuine versus potentially due to chance.

The code creates a boolean mask (True/False table) indicating which p-values meet the Bonferroni threshold, then systematically examines each unique feature pair. By iterating through only the upper triangle of the correlation matrix, we avoid redundant comparisons (e.g., examining both "Feature A vs Feature B" and "Feature B vs Feature A" when they represent the same relationship).

For each feature pair, we extract three key pieces of information: the correlation coefficient (strength and direction of relationship), the p-value (probability the correlation occurred by chance), and a significance determination based on whether the p-value passes the Bonferroni-corrected threshold.

Interpreting significance: A "YES" designation indicates the correlation is statistically significant even after the strict Bonferroni correction - meaning the relationship between those features is real and not due to random chance. Feature pairs marked "YES" with high correlation values (e.g., $|r| > 0.5$) represent potential redundancy concerns, as they measure related information. A "NO" designation means the correlation, while possibly present, is not strong enough to be considered statistically reliable after accounting for multiple testing.

This side-by-side presentation allows us to distinguish between correlations that are both strong and statistically significant (potential redundancy concerns) versus correlations that may appear notable but don't achieve statistical significance after accounting for multiple comparisons.

In [174]...

```
# Create significance mask
significant_bonferroni = p_values_df < bonferroni_alpha

# Display comparison
print("Correlations significant after Bonferroni correction:")
print("(Only showing upper triangle to avoid duplicates)\n")

for i in range(len(relevant_cols)):
    for j in range(i+1, len(relevant_cols)):
        corr_val = corr_focused.iloc[i, j]
        p_val = p_values_df.iloc[i, j]
        sig = "YES" if p_val < bonferroni_alpha else "NO"

        print(f"{relevant_cols[i]} vs {relevant_cols[j]}")
        print(f"    Correlation: {corr_val:.3f}, P-value: {p_val:.6f}, Signifi")
        print()
```

Correlations significant after Bonferroni correction:
(Only showing upper triangle to avoid duplicates)

Weather Condition_encoded vs Track Type_encoded

Correlation: -0.011 , P-value: 0.040132 , Significant: NO

Weather Condition_encoded vs Visibility_encoded

Correlation: -0.062 , P-value: 0.000000 , Significant: YES

Weather Condition_encoded vs Accident Type_encoded

Correlation: 0.016 , P-value: 0.003222 , Significant: NO

Weather Condition_encoded vs RUCC_2023

Correlation: 0.027 , P-value: 0.000001 , Significant: YES

Weather Condition_encoded vs RUCC_Metro_Adjacency_encoded

Correlation: 0.016 , P-value: 0.002731 , Significant: NO

Weather Condition_encoded vs Season_encoded

Correlation: 0.081 , P-value: 0.000000 , Significant: YES

Weather Condition_encoded vs Signalization_encoded

Correlation: 0.024 , P-value: 0.000006 , Significant: YES

Weather Condition_encoded vs Region_encoded

Correlation: -0.066 , P-value: 0.000000 , Significant: YES

Track Type_encoded vs Visibility_encoded

Correlation: -0.054 , P-value: 0.000000 , Significant: YES

Track Type_encoded vs Accident Type_encoded

Correlation: 0.092 , P-value: 0.000000 , Significant: YES

Track Type_encoded vs RUCC_2023

Correlation: -0.228 , P-value: 0.000000 , Significant: YES

Track Type_encoded vs RUCC_Metro_Adjacency_encoded

Correlation: -0.153 , P-value: 0.000000 , Significant: YES

Track Type_encoded vs Season_encoded

Correlation: 0.005 , P-value: 0.325657 , Significant: NO

Track Type_encoded vs Signalization_encoded

Correlation: -0.518 , P-value: 0.000000 , Significant: YES

Track Type_encoded vs Region_encoded

Correlation: -0.059 , P-value: 0.000000 , Significant: YES

Visibility_encoded vs Accident Type_encoded

Correlation: 0.006 , P-value: 0.240852 , Significant: NO

Visibility_encoded vs RUCC_2023

Correlation: 0.047 , P-value: 0.000000 , Significant: YES

Visibility_encoded vs RUCC_Metro_Adjacency_encoded

Correlation: 0.038 , P-value: 0.000000 , Significant: YES

Visibility_encoded vs Season_encoded
Correlation: -0.020, P-value: 0.000161, Significant: YES

Visibility_encoded vs Signalization_encoded
Correlation: 0.041, P-value: 0.000000, Significant: YES

Visibility_encoded vs Region_encoded
Correlation: 0.010, P-value: 0.065006, Significant: NO

Accident Type_encoded vs RUCC_2023
Correlation: -0.094, P-value: 0.000000, Significant: YES

Accident Type_encoded vs RUCC_Metro_Adjacency_encoded
Correlation: -0.072, P-value: 0.000000, Significant: YES

Accident Type_encoded vs Season_encoded
Correlation: -0.020, P-value: 0.000169, Significant: YES

Accident Type_encoded vs Signalization_encoded
Correlation: 0.032, P-value: 0.000000, Significant: YES

Accident Type_encoded vs Region_encoded
Correlation: -0.014, P-value: 0.009652, Significant: NO

RUCC_2023 vs RUCC_Metro_Adjacency_encoded
Correlation: 0.868, P-value: 0.000000, Significant: YES

RUCC_2023 vs Season_encoded
Correlation: 0.001, P-value: 0.834086, Significant: NO

RUCC_2023 vs Signalization_encoded
Correlation: 0.092, P-value: 0.000000, Significant: YES

RUCC_2023 vs Region_encoded
Correlation: -0.044, P-value: 0.000000, Significant: YES

RUCC_Metro_Adjacency_encoded vs Season_encoded
Correlation: 0.002, P-value: 0.710264, Significant: NO

RUCC_Metro_Adjacency_encoded vs Signalization_encoded
Correlation: 0.055, P-value: 0.000000, Significant: YES

RUCC_Metro_Adjacency_encoded vs Region_encoded
Correlation: -0.069, P-value: 0.000000, Significant: YES

Season_encoded vs Signalization_encoded
Correlation: 0.002, P-value: 0.749515, Significant: NO

Season_encoded vs Region_encoded
Correlation: -0.014, P-value: 0.007206, Significant: NO

Signalization_encoded vs Region_encoded
Correlation: 0.067, P-value: 0.000000, Significant: YES

Summary: Pearson Correlation and Bonferroni Correction Findings

Analysis of pairwise feature correlations with Bonferroni-corrected significance testing ($\alpha = 0.00139$ for 36 tests) revealed several key patterns in feature relationships:

Strong redundancy identified:

- Visibility and Time_of_Day show high negative correlation (-0.676 , $p < 0.001$), indicating these features measure essentially the same underlying information (lighting conditions vary predictably with time). Only one should be retained to avoid multicollinearity.
- Track Type and Signalization demonstrate moderate negative correlation (-0.518 , $p < 0.001$), reflecting that yard tracks are typically unsignalized while mainline tracks are signalized. These features contain overlapping information.

Moderate correlations:

- Track Type shows significant correlations with Equipment Type (0.273), RUCC codes (-0.228), and multiple other features, suggesting it captures operational context that intersects with several dimensions.
- Equipment Type correlates moderately with RUCC (-0.173) and Signalization (-0.224), reflecting differences between urban and rural operations.

Independent features:

- Day of Week shows no statistically significant correlations with any other features after Bonferroni correction, providing unique temporal information.
- Weather Condition and Accident Type show only weak correlations with other features, suggesting they contribute independent predictive information.

These findings indicate that from our nine candidate features, we should select one feature from each highly correlated pair (Visibility/Time_of_Day and Track Type/Signalization) while retaining the independent features. Final feature selection will also consider each feature's individual predictive power for accident cause categories through chi-square testing.

Chi-Square Independence Tests with Bonferroni Correction

Test whether each candidate feature is statistically associated with accident Cause_Category using chi-square tests of independence.

Chi-square tests evaluate whether the distribution of cause categories differs significantly across the levels of each feature. A significant result indicates that knowing a feature's value provides information about the likely accident cause, making it a potentially useful predictor for classification.

Since we're conducting multiple tests (one for each of the 9 features), we apply Bonferroni correction by dividing the standard significance level (0.05) by the number of tests. This adjusted threshold controls for the increased risk of false positives when performing multiple hypothesis tests.

For each feature, we calculate the chi-square statistic, p-value, and degrees of freedom, then compare both the raw p-value ($\alpha = 0.05$) and Bonferroni-corrected p-value to determine significance. Features that remain significant after Bonferroni correction demonstrate robust associations with accident causes and are strong candidates for inclusion in classification models.

Interpreting results: The chi-square statistic measures the magnitude of association between each feature and accident causes - higher values indicate stronger relationships. P-values indicate the probability of observing such an association by random chance. Features with p-values below the Bonferroni threshold demonstrate statistically significant associations that are unlikely to be false discoveries. The "Sig_Bonferroni" column identifies features that pass this conservative threshold and represent the most reliable predictors. Combined with the correlation analysis, these results guide final feature selection by balancing predictive power (chi-square s

```
In [164... from scipy.stats import chi2_contingency

# Chi-square test for each feature vs Cause_Category
chi_square_results = []

for feature in relevant_cols:
    contingency = pd.crosstab(df_encoded[feature], df_encoded['Cause_Category'])
    chi2, p_val, dof, expected = chi2_contingency(contingency)

    chi_square_results.append({
        'Feature': feature,
        'Chi2': chi2,
        'P-value': p_val,
        'DOF': dof
    })

chi_results_df = pd.DataFrame(chi_square_results)

n_tests = len(relevant_cols)
bonferroni_alpha_chi = 0.05 / n_tests

chi_results_df['Sig_raw'] = chi_results_df['P-value'] < 0.05
chi_results_df['Sig_Bonferroni'] = chi_results_df['P-value'] < bonferroni_alpha_chi
```



```
print(f"Bonferroni corrected alpha: {bonferroni_alpha_chi:.6f}\n")
print(chi_results_df.to_string(index=False))
```

Bonferroni corrected alpha: 0.005556

	Feature	Chi2	P-value	D0F	Sig_raw	Sig_Bonf
erroni						
Weather	Condition_encoded	221.901534	5.686543e-34	24	True	
True						
	Track Type_encoded	6191.808652	0.000000e+00	16	True	
True						
	Visibility_encoded	262.358092	1.499365e-46	16	True	
True						
	Accident Type_encoded	21483.325831	0.000000e+00	52	True	
True						
	RUCC_2023	740.735930	3.820692e-135	32	True	
True						
	Equipment_Type_encoded	6801.284558	0.000000e+00	56	True	
True						
	Day_of_Week_encoded	87.514228	3.702854e-09	24	True	
True						
	Signalization_encoded	4533.414499	0.000000e+00	8	True	
True						
	Region_encoded	217.915118	6.421600e-40	12	True	
True						

Summary: Chi-Square Independence Test Findings

Chi-square tests with Bonferroni correction ($\alpha = 0.00556$ for 9 tests) evaluated whether each feature for consideration shows statistically significant associations with accident Cause_Category.

Key findings: All nine candidate features demonstrated statistically significant associations with accident causes, with p-values well below the Bonferroni-corrected threshold. This indicates that each feature provides genuine predictive information about accident cause categories.

Features with strongest associations (highest chi-square statistics):

- Accident Type: Extremely strong association with cause categories
- Equipment Type: Very strong predictor of accident causes
- Track Type and Signalization: Both show very strong associations
- RUCC codes: Strong association confirming rural/urban patterns differ by cause

Features with moderate associations:

- Time of Day and Visibility: Both significant predictors
- Weather Condition: Significant but more modest association
- Day of Week: Weakest association among significant features, but still passes Bonferroni correction

Implications for feature selection: While all features are statistically significant predictors, the correlation analysis revealed redundancy between certain pairs (Visibility/Time_of_Day and Track Type/Signalization). Since both members of these pairs are significant, the choice of which to retain should be based on interpretability, data quality, and team discussion. Features showing both strong predictive power and independence from other features (Equipment Type, RUCC, Accident Type, Weather Condition, Day of Week) are clear candidates for inclusion in classification models.

```
In [165... df_before = len(df_merged)
nulls = df_merged['Cause_Category'].isna().sum()

print(f"Before: {df_before}")
print(f"Nulls in Cause_Category: {nulls}")

df_merged = df_merged.dropna(subset=['Cause_Category'])

print(f"After: {len(df_merged)}")
print(f"Dropped: {df_before - len(df_merged)}")
```

```
Before: 34562
Nulls in Cause_Category: 2
After: 34560
Dropped: 2
```

```
In [166... cols_to_check = ['Cause_Category', 'Weather Condition', 'Track Type', 'Visib

df_before = len(df_merged)
nulls_per_col = df_merged[cols_to_check].isna().sum()

print(f"Before: {df_before}")
print(f"Nulls per column:\n{nulls_per_col}")

df_merged = df_merged.dropna(subset=cols_to_check)

print(f"After: {len(df_merged)}")
print(f"Dropped: {df_before - len(df_merged)}")
```

```
Before: 34560
Nulls per column:
Cause_Category      0
Weather Condition   4
Track Type          2
Visibility           8
Accident Type       1
Region              0
RUCC_2023           0
dtype: int64
After: 34551
Dropped: 9
```

```
In [167... ## Final Step: Create Train/Val/Test Splits
import os
# Create splits
from sklearn.model_selection import train_test_split
```

```

# First split: 80/20 (working/test)
df_working, df_test = train_test_split(
    df_merged,
    test_size=0.2,
    stratify=df_merged['Cause_Category'],
    random_state=RANDOM_SEED
)

# Second split: 75/25 of working = 60/20 (train/val) of total
df_train, df_val = train_test_split(
    df_working,
    test_size=0.25,
    stratify=df_working['Cause_Category'],
    random_state=RANDOM_SEED
)

# Save
df_train.to_csv('../data/splits/train.csv', index=False)
df_val.to_csv('../data/splits/validate.csv', index=False)
df_test.to_csv('../data/splits/test.csv', index=False)

print(f"Train: {len(df_train)}")
print(f"Val: {len(df_val)}")
print(f"Test: {len(df_test)}")

```

Train: 20730

Val: 6910

Test: 6911

Phase 2: Cause Code EDA - Summary

What We Did

Starting from the cleaned merged dataframe produced in Phase 1, we explored features to understand which ones show meaningful differences across the five accident cause categories (H, M, T, E, S).

1. **Correlation Matrix** - Checked relationships between numerical features to identify redundancy. Found Track Type/Train Speed and Population/RUCC are partially redundant pairs.
2. **Track Type vs Cause Category** - Mainline tracks are dominated by Mechanical failures, while Yards and Industry tracks show higher Human Factor rates. Strong signal for classification.
3. **RUCC Code vs Cause Category** - Metro counties (1-3) are dominated by Human Factor causes, while rural counties (7-9) show increasing Track and Mechanical failures. Visually confirms Phase 1's statistical findings.

4. **Accident Month vs Cause Category** - Cause proportions remain stable across all 12 months. Month does not appear to be a strong predictor - useful to know what to deprioritize.
5. **Train Composition (pct_empty)** - Engineered feature measuring percentage of empty cars in consist. Track and Mechanical accidents occur with heavier trains (lower pct_empty), while Human Factor accidents occur with lighter consists (higher pct_empty). ANOVA confirmed statistical significance ($p < 0.001$). Shows moderate correlation with Track Type and RUCC, providing complementary information.

Key Takeaways

- **Track Type, RUCC Code, and pct_empty** are the strongest candidates for cause code classification so far
- **Accident Month** can likely be deprioritized
- Due to redundancy, we likely only need one of Population/RUCC and one of Track Type/Train Speed going forward

Possible Next Steps

- Feature engineering: Create US Census region from state codes to capture broader geographic patterns beyond RUCC
- Begin feature selection - narrow down to the most useful features before moving into classification
- Coordinate with the team on shared preprocessing decisions (e.g., handling the nan values in encoded columns)
- Consider a simple baseline classifier (e.g., Logistic Regression) to establish a benchmark before trying more complex models